

TEXAS A&M UNIVERSITY

ROBOTICS TEAM & LEADERSHIP EXPERIENCE

Statics/Solids Technical Interview Concepts

Week 5

Attendance

Intro

- Mechanical engineers are tasked with very important jobs when entering the workforce
- Employers need to know that candidates have a fundamental understanding of certain concepts
- The questions companies ask depend on their field
- Such fields include oil and gas, HVAC, aerospace, energy, construction, industrial machining, manufacturing, robotics, automotive, etc…

Concepts Covered Today

- FBD
- Inertia
- Shear and Moment
- Beam Deflection
- Stress and Strain
- Von Mises
- Buckling
- Stiffness
- Vibration
- Thermo
- Fluids



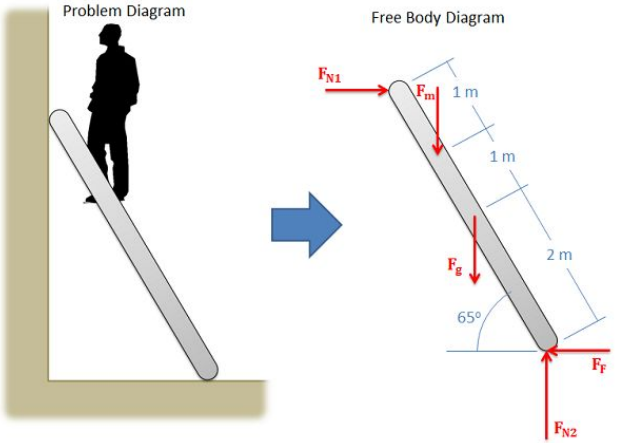
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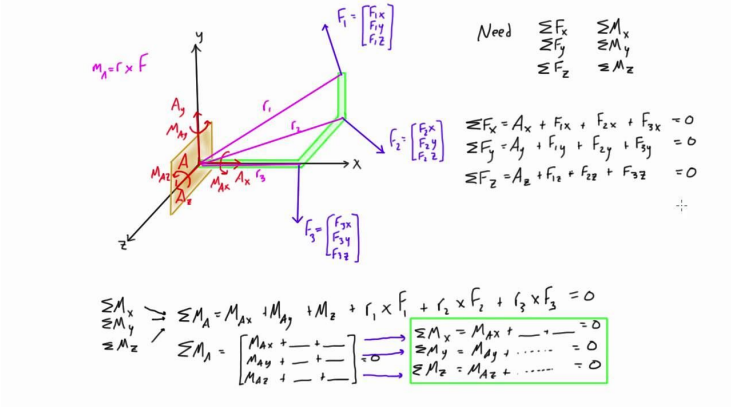
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Free Body Diagrams

Free Body Diagrams (FBDs)

- One of the most fundamental concepts that engineers learn and use
- FBDs simplify systems and visualize forces, moments, stresses, etc. acting on a body

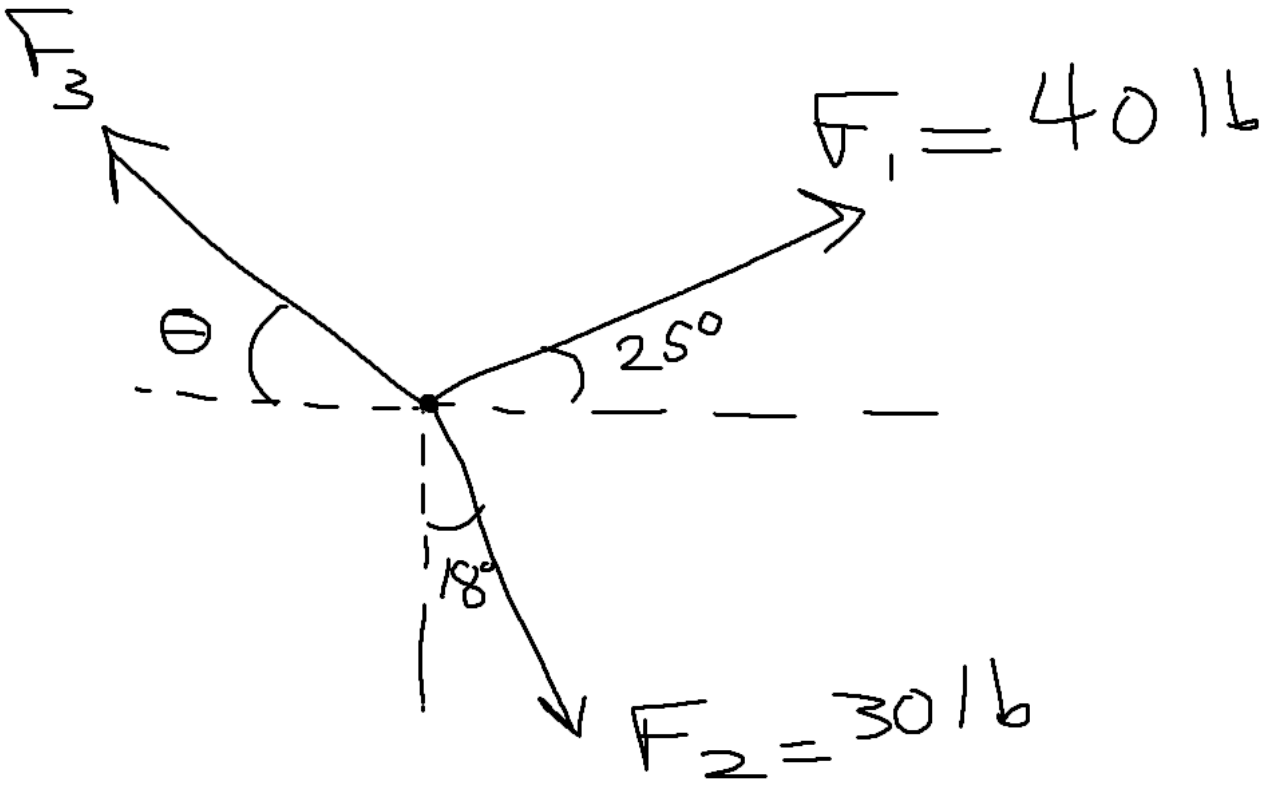




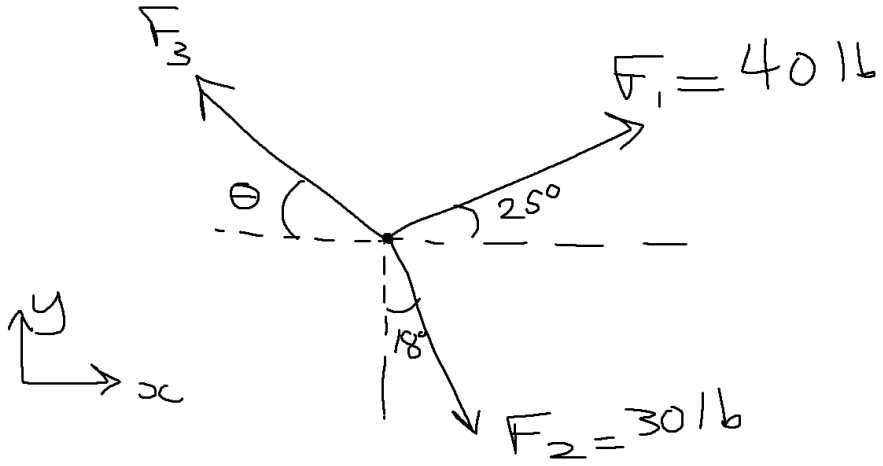
FBD Basics

- Step 1: MAKE A CONSISTENT AXIS SIGN CONVENTION
- Step 2: Label every force and moment
- Step 3: Identify unknowns
- Step 4: Perform sum of forces and moments analyses
  - Remember right hand rule!
- Step 5: Solve for unknowns
- Step 6: CHECK UNITS

FBD Example



FBD Example Solution



$\Sigma F_x = 0 = 40\cos 25 + 30\sin 18 - F_3\cos\theta$

$F_3 = \dfrac{45.5}{\cos\theta}$

$\Sigma F_y = 0 = 40\sin 25 - 30\cos 18 + F_3\sin\theta$

$F_3 = \dfrac{11.6}{\sin\theta}$

$\dfrac{\sin\theta}{\cos\theta} = \dfrac{11.6}{45.5}$

$\theta = 14.3°$

$F_3 = 12$ lb



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Inertia

Inertia



- Inertia is an **object's natural tendency to resist any change in its state of motion**
  - Newton's 1st Law: An object at rest stays at rest. An object in motion stays in motion unless acted upon by an outside force.

- For example, when a car travelling 60 mph stops to a halt, you lurch forward because your body is still travelling at 60 mph.

- 2 Types: **Mass MOI and Area MOI,** *however we shall focus on Area MOI for example problem cause 225*

- **Parallel Axis Theorem:** allows you to find the MOI of an object around an axis using its center of mass, mass, and distance
- PAT Basic Formula:

$$I = I_{cm} + md^2$$

Mass MOI

- Mass MOI: **measures an object's resistance to rotation**.
- Shows how hard it is to spin an object or change spinning speed
- Units: kg * m^2 or lb * (ft*s^2)
- Formula:

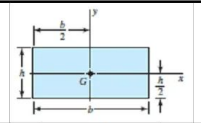
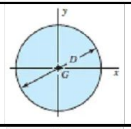
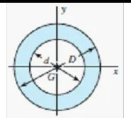
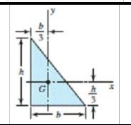
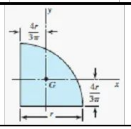
$$I = \int_m r^2 dm$$

Area MOI

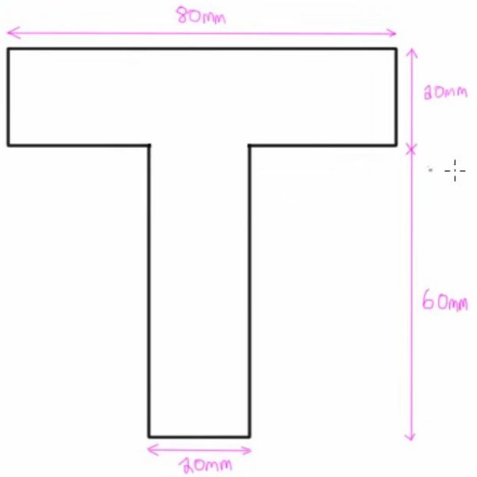
- Area MOI: **measures a shape's resistance to bending**
- Shows how well a structural cross section like a beam resists bending forces
- Units: m^4 or ft^4
- Formula:

$$I_x = \int y^2 dA$$

$$I_y = \int x^2 dA$$

Cross Section		Cross-Sectional Area (A)	Major Axis Moment of Area (I_x)	Minor Axis Moment of Area (I_y)
Rectangle		bh	$bh^3/12$	$b^3h/12$
Circle		$\pi D^2/4$	$\pi D^4/64$	$\pi D^4/64$
Circular Tube		$\pi/4(D^2 - d^2)$	$\pi/64(D^4 - d^4)$	$\pi/64(D^4 - d^4)$
Right Triangle		$bh/2$	$bh^3/36$	$b^3h/36$
Quarter Circle		$\pi r^2/4$	$r^4(\pi/16 - 4/(9\pi))$	$r^4(\pi/16 - 4/(9\pi))$

Area MOI Example



Solve for I_x

How to Solve:

1. Break model into simpler shapes
2. Determine centroid of shape with respect to x and y
3. Label shapes A & B
4. Make table for Area, $\bar{Y}$, $\bar{Y}A$
5. Find A, $\bar{Y}$, and $\bar{Y}A$ for A & B
6. Find $\bar{Y}$ for combined
7. Solve for Inertia

Formulas:

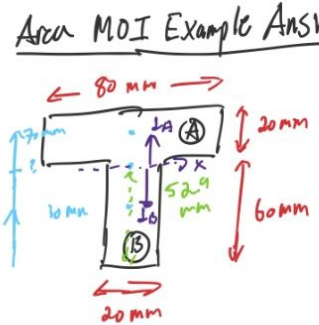
$$\bar{Y} = \frac{\sum_i \bar{Y}A}{\sum_i A}$$

$$I = \frac{1}{12}bh^3$$

$$I_x = \sum_i \left(I_i + Ad_i^2\right)$$

	A	$\bar{Y}$	$\bar{Y}A$
A			
B			
Σ:			Σ:

Area MOI Example Answers



Tips!!

- Parallel Axis Theorem
- split into familiar shapes
- Guess Centroid location by verifying for each shape + entire model

- Construct Centroid table

	Area	$\bar{y}$	$\bar{y}A$
A	$1600\,mm^2$	70 mm	$112\,000\ mm^3$
B	$1200\,mm^2$	30 mm	$36,000\ mm^3$
Σ	$2800\,mm^2$		$148\,000\ mm^3$

$$\bar{y} = \frac{\sum_i \bar{y}A}{\sum_i A} = \frac{148\,000\ mm^3}{2800\ mm^2} = 52.9\ mm$$

$$I = \frac{1}{12}bh^3$$

$$I_x = \sum_i (I_i + A_i d_i^2) = \sum_i \left(\frac{1}{12}bh^2 + Ad^2\right) = \frac{1}{12}(80\ mm)(20\ mm)^3 + (80)(20)(70-53)^2$$

$$+ \frac{1}{12}(200)(60)^3 + (200)(60)(53-30)^2$$

$$\Rightarrow 1,510,533\ mm^4 \Rightarrow 1511 \times 10^{-9}\ m^4$$



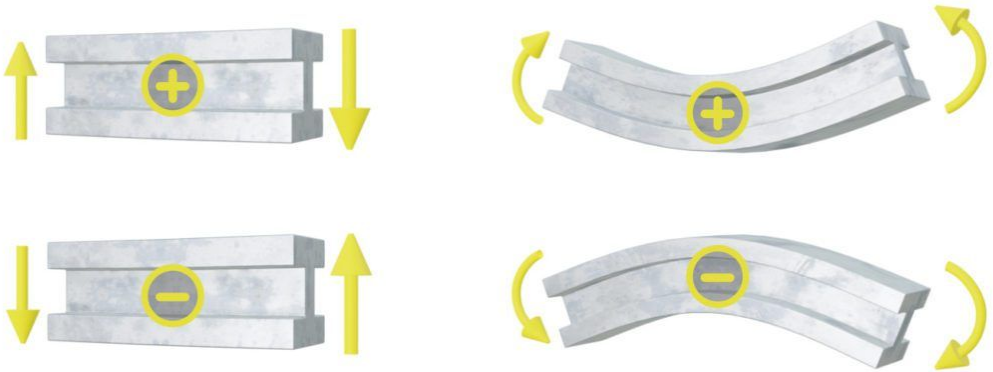
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Internal Forces

Shear and Bending Moment

- **Shear (V)**: internal force acting parallel to the cross section of the beam
  - Shear tries to slice the parts perpendicular to the axis of the beam
- **Bending Moment (M)**: internal rotational force causing tension and compression within the beam
  - BM curves the member due to rotational effect



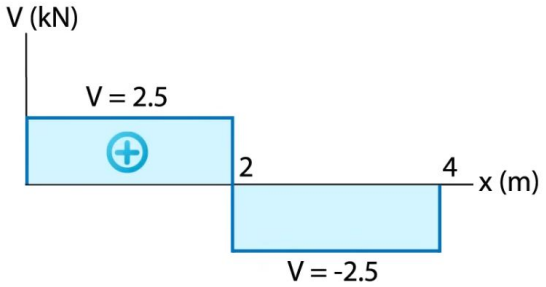
Shear and Bending Moment

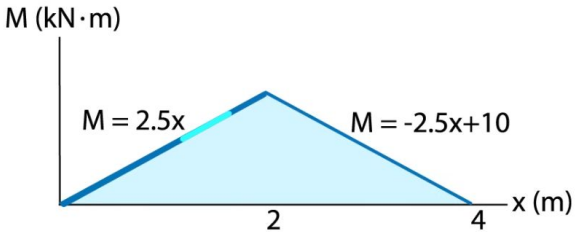
- Positive sign convention for shear creates a clockwise rotation about the beam, negative convention would be CCW
- Positive sign convention for moment is concave up (smiling)

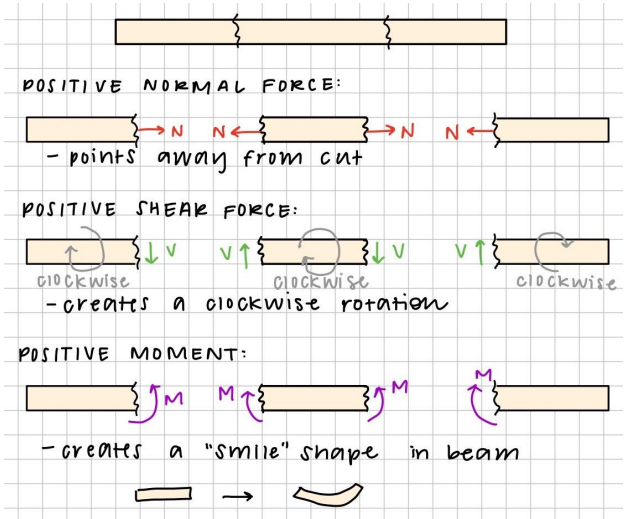
Shear Units: kN
Moment Units: kN*m

$$\Delta M = \int V \, dx$$

$$\frac{dM}{dx} = V$$







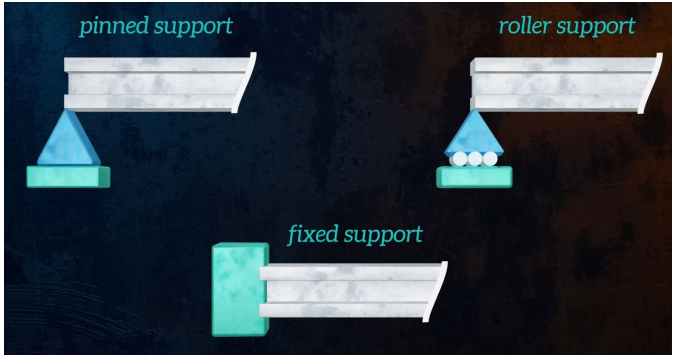
Shear and Bending Moment Important Key Terms

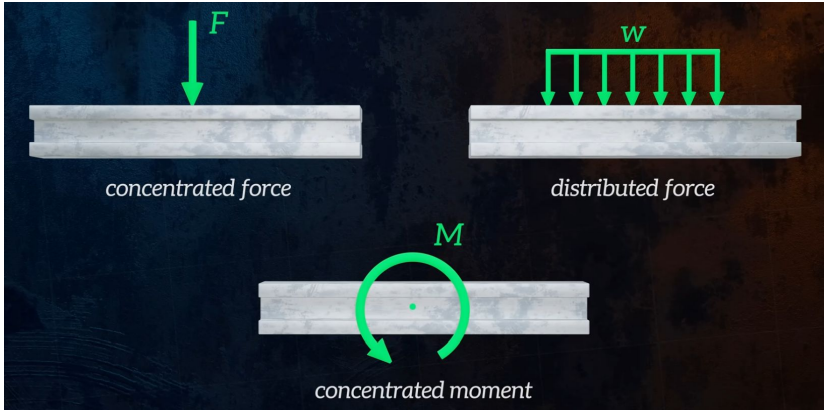
Forces

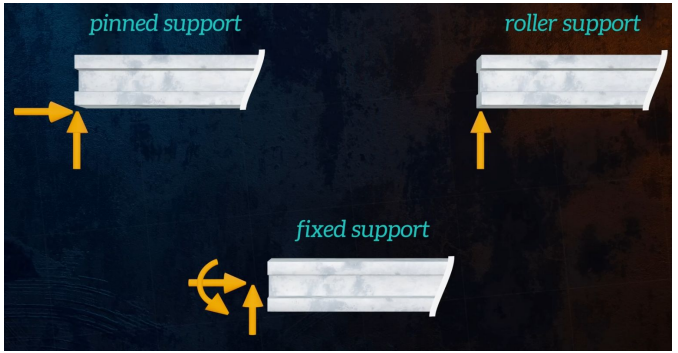
- **Concentrated Force (N)**: force applied at a single point
- **Distributed Force (N/m)**: force spread continuously over a length
- **Concentrated Moment**: rotational load applied at a point, causes a jump in moment diagram, no effect on shear

Supports

- **Pinned Support**: allows rotation but restrains movement in x or y direction, **2 reaction forces**, drawn as a triangle
- **Roller Support**: allows rotation and horizontal sliding, **1 reaction force**, triangle with circles underneath
- **Fixed Support**: restrains everything, **2 reaction forces and reaction moment**, drawn as a hatched wall

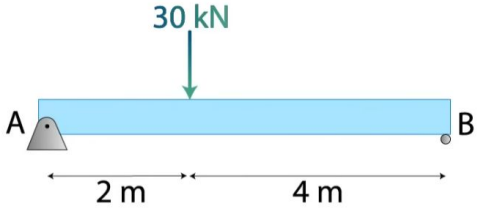






Shear and Moment Example

Draw the shear and moment diagrams for the beam.



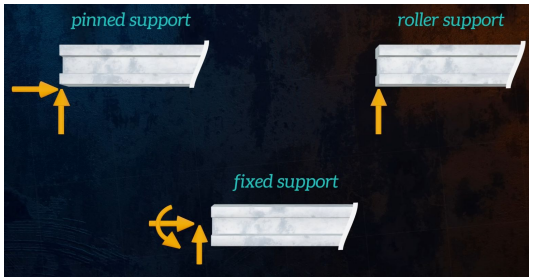
Steps to Solve:

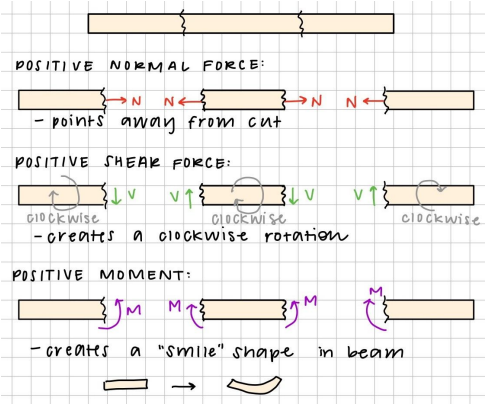
1. Draw FBD, remember reaction forces
2. Equations of equilibrium
3. Solve for Ax, Ay, and By
4. Split beam into respective segments
5. Find shear and moment for both segments
6. Draw diagrams

Formulas:

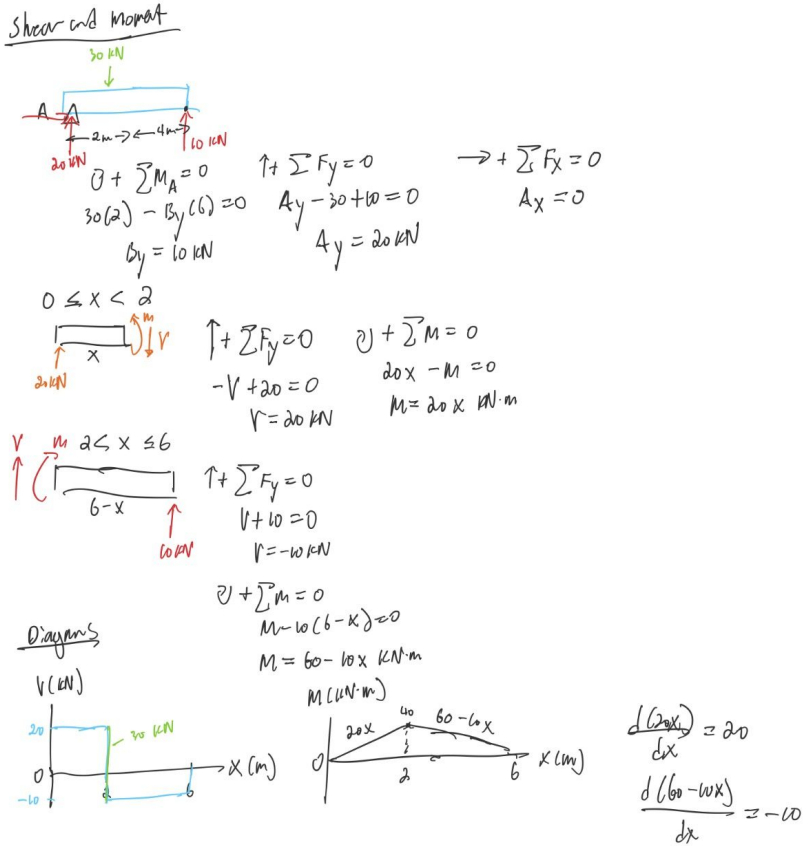
$$\Delta M = \int V \, dx$$

$$\frac{dM}{dx} = V$$





Shear and Moment Example Answers





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Stress and Strain

Stress and Strain

- Stress: internal resisting force per unit area



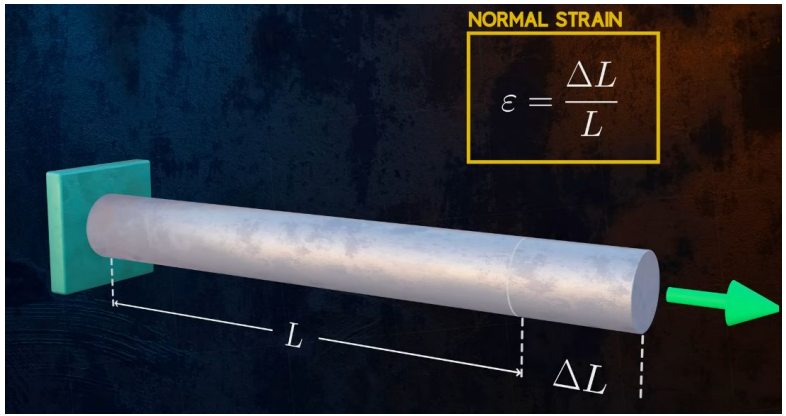
$$\sigma = \frac{F}{A}$$

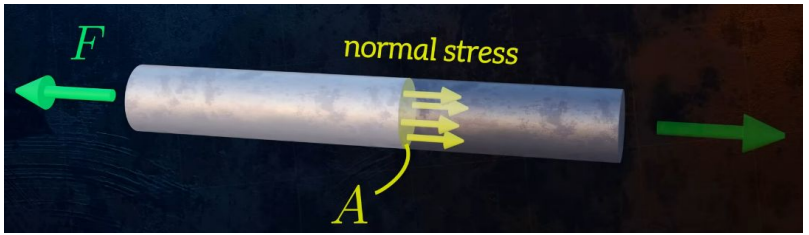
Units: Pascals (Pa) or N/m^2

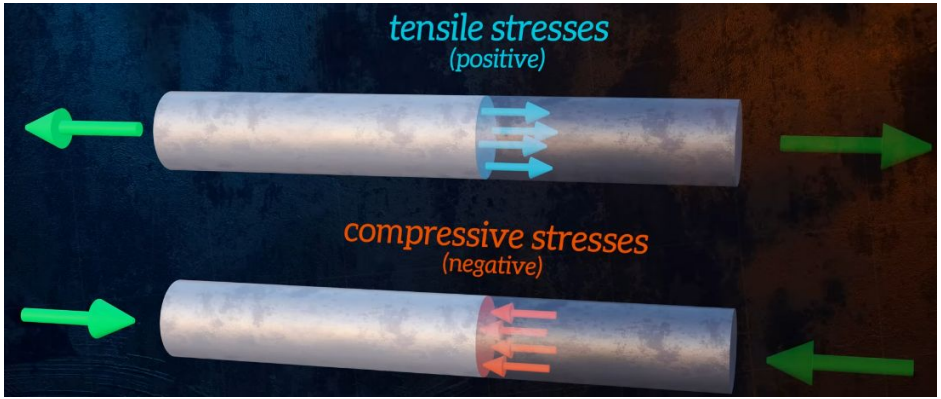
- Strain: resulting physical change in displacement or deformation in materials shape or size

$$\varepsilon = \frac{\Delta L}{L_0}$$

Units: Unitless, change in length over length initial





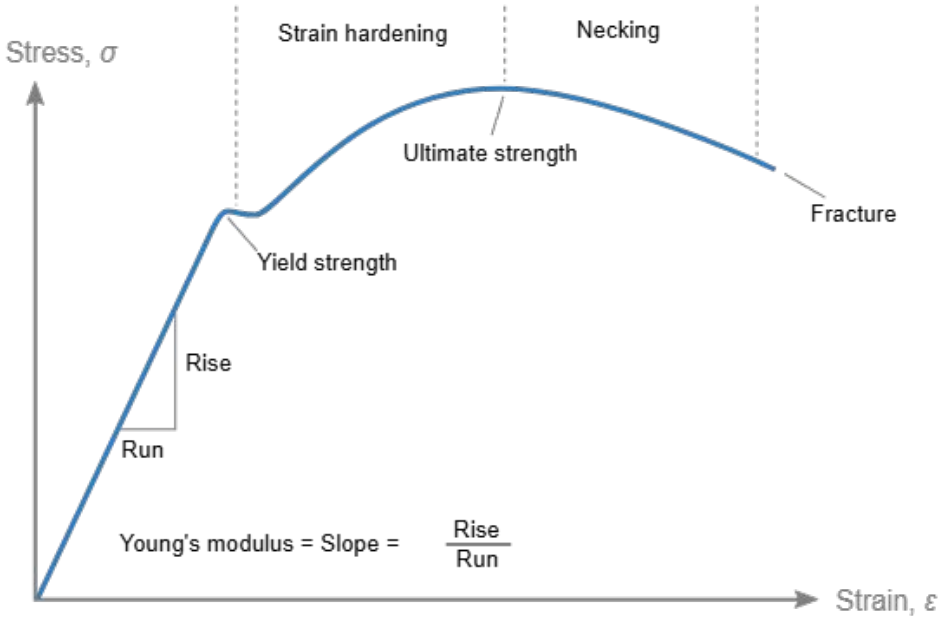


- Stress and strain can be both positive and negative
- Tensile is typically positive and compressive is typically negative

Stress and Strain Curve

Stress and Strain Curve Key Terms

- **Elastic Modulus (E)** is the slope of the elastic deformation (not permanent)
- **Yield strength** is the maximum stress a component can endure before permanent deformation
- **Strain hardening** is when a material undergoes plastic deformation and gets stronger as a result
- **Ultimate tensile strength (UTS)** is the maximum tensile strength, the highest point on the graph
- **Necking** is when a ductile material reaches UTS and begins to decrease in cross sectional area
- **Fracture** is when the material breaks and snaps





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Beam Deflection

Beam Deflection

- Beam Deflection is the **displacement** of a beam from its original position due to loading
- Find how much the beam is bending, aka **Beam Deflection** δ_{max}
- P is the load/force in N
- L is length of the beam in m
- E is elastic modulus in GPa
- I is inertia

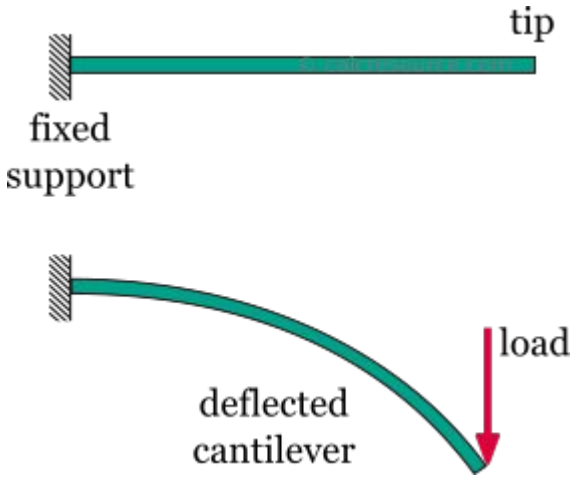
$$\frac{d^2v}{dx^2} = \frac{M(x)}{EI}$$

$$I = \frac{bh^3}{12}$$

$$\delta_{max} = \frac{PL^3}{3EI}$$

Beam Deflection

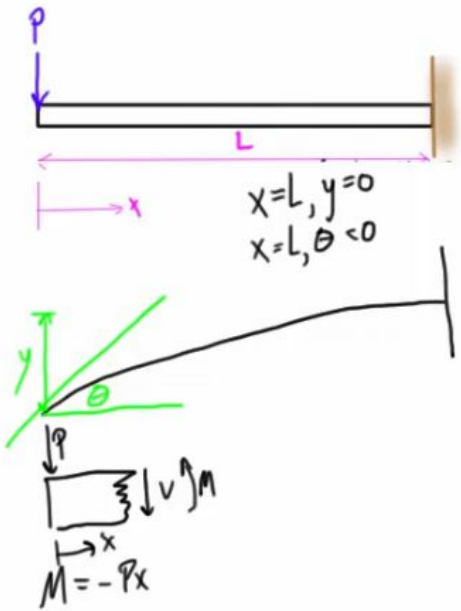
- Common example in statics: Cantilever Beam
- Great visualization of shear force, bending moment, and vertical load
- How to Approach?
  - Write Assumptions
  - Free Body Diagram
  - Set Equations of Equilibrium
  - Solve



Beam Deflection

- **Derive the Maximum Beam Deflection and Slope of a cantilever beam with a point load**

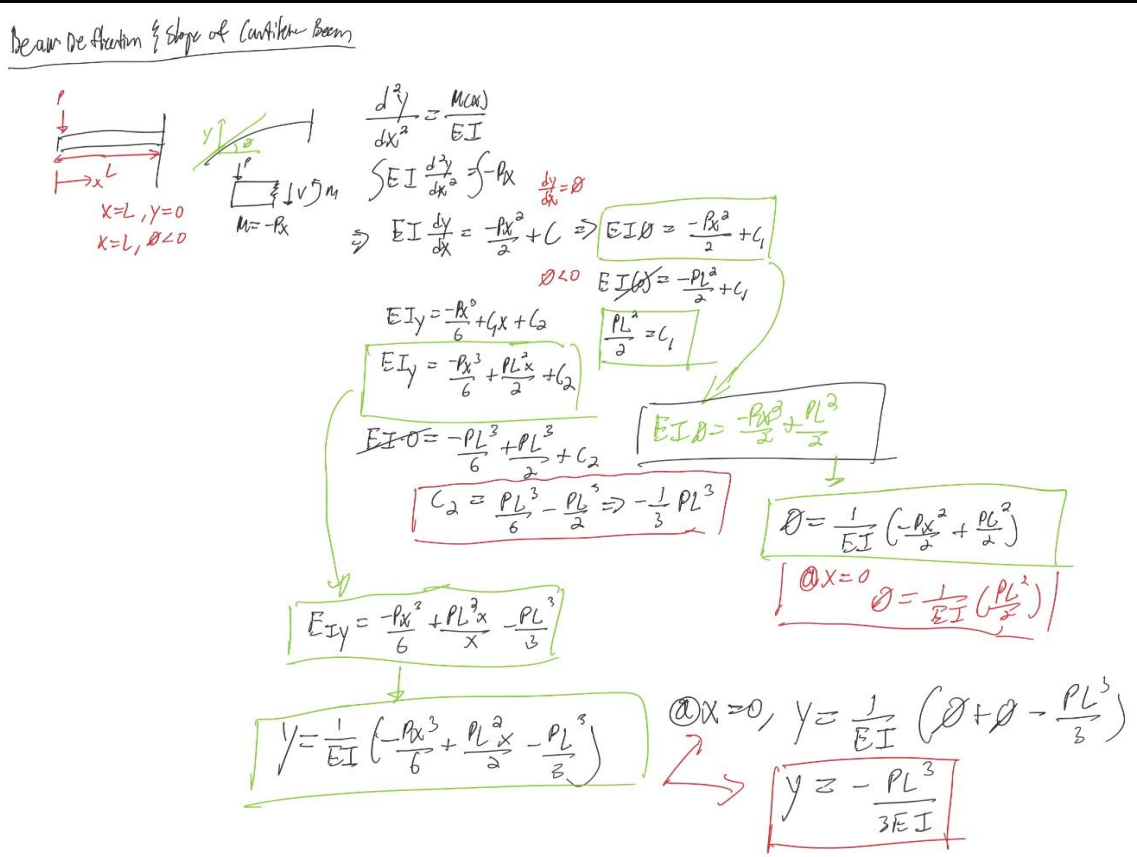
Hint: You will need to use double integration



$$\frac{d^2y}{dx^2} = \frac{M(x)}{EI}$$

$$EI\frac{d^2y}{dx^2} = -Px$$

Beam Deflection Solution





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Buckling

Buckling

- Too much compressive load on a beam
- Causes beam to bend outwards and lose stability, results in large displacements
  - Imagine pushing on both ends of a ruler, it starts to bend outward towards the middle



Buckling

- Fundamental formula to calculate buckling load
- Used for slender column
- Euler's Buckling Formula:

$$P_{cr} = \frac{\pi^2 EI}{(KL)^2}$$

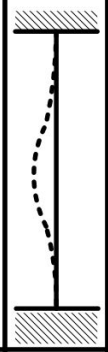
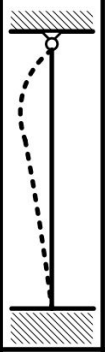
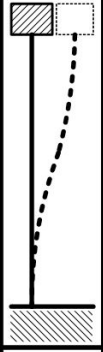
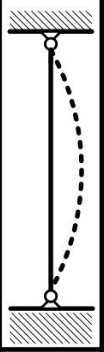
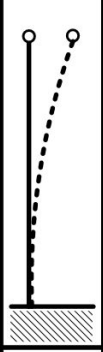
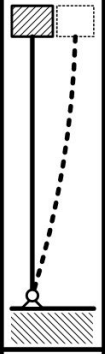
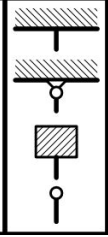
- **Pcr = Buckling Load**
- **E = Young's Modulus** - measure of the stiffness of material which is the resistance to stretching/compressing
- **I = Minimum Area of Inertia**
- **K= Column Effective Length Factor** - multiplier that finds columns effective buckling length (unitless)
- **L = Length**

Buckling

- Stress concentration Factor (K) have a theoretical and recommended design value

From end to end:

- **Pinned - Pinned:** K = 1.0 for both theoretical and experimental
- **Fixed - Fixed:** K = 0.65 for theoretical and 0.5 for experimental
- **Fixed - Pinned:** K = 0.8 for theoretical and 0.7 for experimental
- **Fixed - Free:** K = 2.1 theoretical and 2.0 for experimental

Buckled shape of column shown by dashed line						
Theoretical K value	0.5	0.7	1.0	1.0	2.0	2.0
Recommended design value K	0.65	0.80	1.2	1.0	2.10	2.0
End condition key		Rotation fixed and translation fixed Rotation free and translation fixed Rotation fixed and translation free Rotation free and translation free				

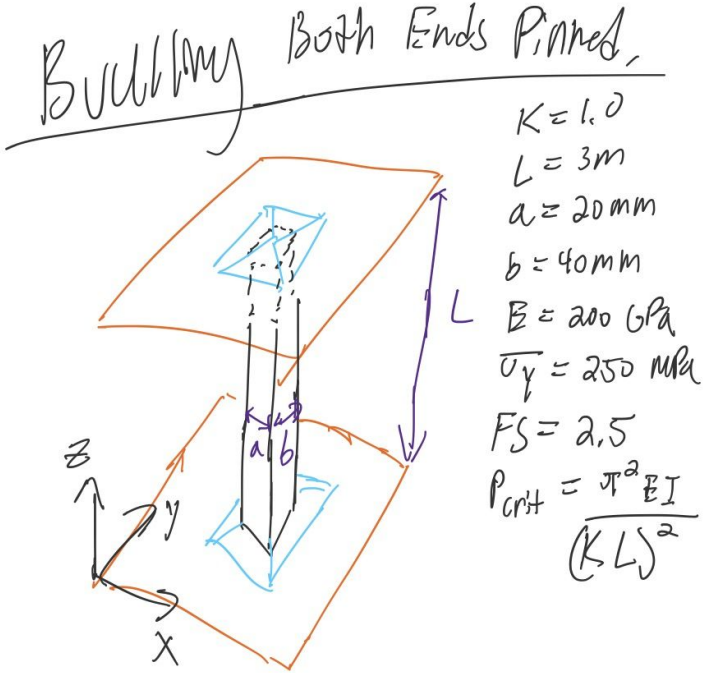
Buckling

Factor of Safety: design multiplier that is used to ensure that a component can handle significantly more load/stress than expected during normal operation

- **FoS = 1:** structure operates at absolute limit, any extra load causes failure
- **FoS > 1:** structure operates beyonds normal use and is safe
- **FoS < 1:** load exceeds material strength and will cause failure



How to Solve Buckling



- Both Ends Pinned, K = 1.0
- Find Pcrit in kN, the critical load with a FoS of 2.5.
- Find the allowable load in kN

$$FS = \frac{\text{Ultimate Capacity (or Stress/Load)}}{\text{Allowable Capacity (or Stress/Load)}}$$

Buckling Solution

$$I_y = \frac{1}{12} b a^3 = \frac{1}{12}(0.04)(0.02)^3$$

$$\Rightarrow 0.0267 \times 10^{-6}\ m^4$$

$$I_x = \frac{1}{12} a b^3 = \frac{1}{12}(0.02)(0.04)^3$$

$$\Rightarrow 0.1067 \times 10^{-6}\ m^4$$

$$P_{crit_{yz}} = \frac{\pi^2 E I_y}{K \cdot L^2} \Rightarrow \frac{\pi^2 \cdot 200 \times 10^9\ N/m^2 \cdot 0.0267 \cdot 10^{-6}\ m^4}{1 \cdot (3m)^2} = 5.9\ kN$$

$$P_{crit_{xz}} = \frac{\pi^2 E I_x}{K L^2} \Rightarrow \frac{\pi^2 \cdot 200 \times 10^9 \cdot 0.1067}{3^2} = 23.4\ kN$$

$$FS = \sigma_{allow} \rightarrow all = \frac{5.9\ kN}{2.5} = 2.36\ kN$$

$\sigma_y = 250\ MPa = P/A$ area $= 800\ mm^2$

$\rightarrow P_y = 250\ MPa \cdot 800\ mm^2 \Rightarrow 200\ kN$

$$allow = \frac{200\ kN}{2.5} = 80\ kN$$



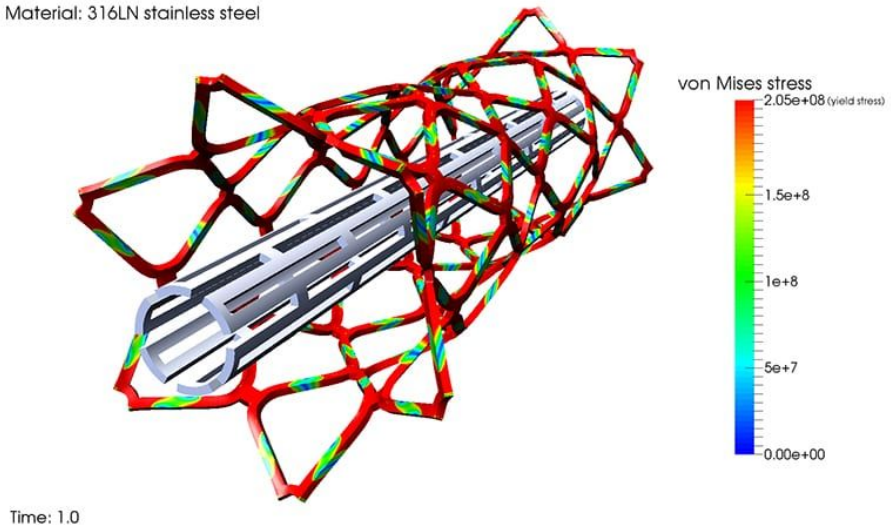
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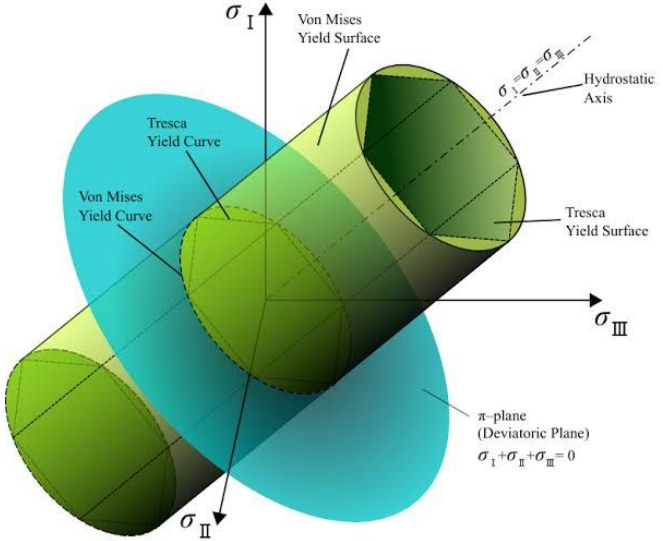
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Von Mises Stress

Von Mises Stress

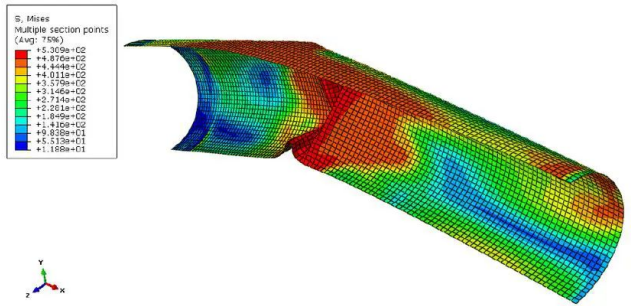
- Von Mises Stress is a scalar value used by engineers to determine if a ductile material will begin to yield or deform under a complex/multi-axial load state





Von Mises Stress

- Simply put, Von Mises is the test of a material's ability against chaos.
- During FEA (Finite Element Analysis) simulations, von mises stresses are the most common and apply twisting, bending, and shearing all at once
- It gives great data on what areas on the model fail first, typically indicated by bright red



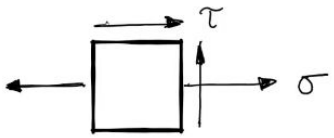
Von Mises Stress Example

- 2-D Example
- 3-D is pain

Von Mises Stress for a Given $\boldsymbol{\sigma}$ & $\boldsymbol{\tau}$

TORSION & SHEAR

AXIAL LOADING & BENDING



MOHR'S CIRCLE :

CENTER : $\frac{\sigma+0}{2} = \frac{\sigma}{2}$

RADIUS : $R = \sqrt{\left(\frac{\sigma-0}{2}\right)^2 + \tau^2}$

$\sigma_1 = \frac{\sigma}{2} + R$

$\sigma_2 = 0$

$\sigma_3 = \frac{\sigma}{2} - R$

$$\sigma' = \left[\frac{(\sigma_1-\sigma_2)^2 + (\sigma_2-\sigma_3)^2 + (\sigma_3-\sigma_1)^2}{2}\right]^{1/2}$$

Von Mises Stress Solution

$$\sigma' = \left[\frac{(\sigma_1-\sigma_2)^2+(\sigma_2-\sigma_3)^2+(\sigma_3-\sigma_1)^2}{2}\right]^{1/2}$$

$$\sigma' = \left[\frac{\left(\frac{\sigma}{2}+R\right)^2+\left(-\left(\frac{\sigma}{2}-R\right)\right)^2+\left(\left(\frac{\sigma}{2}-R\right)-\left(\frac{\sigma}{2}+R\right)\right)^2}{2}\right]^{1/2}$$

$$= \left[\frac{\left(\left(\frac{\sigma}{2}\right)^2+2\frac{\sigma}{2}R+R^2\right)+\left(\left(\frac{\sigma}{2}\right)^2-2\frac{\sigma}{2}R+R^2\right)+(-2R)^2}{2}\right]^{1/2}$$

$$= \left[\frac{2\left(\frac{\sigma^2}{4}\right)+2R^2+4R^2}{2}\right]^{1/2} = \left[\frac{\sigma^2}{4}+3R^2\right]^{1/2}$$

$$= \left[\frac{\sigma^2}{4}+3\left(\left(\frac{\sigma}{2}\right)^2+\tau^2\right)\right]^{1/2} = \left[\frac{\sigma^2}{4}+\frac{3\sigma^2}{4}+3\tau^2\right]^{1/2} = \sqrt{\sigma^2+3\tau^2}$$



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Vibrations

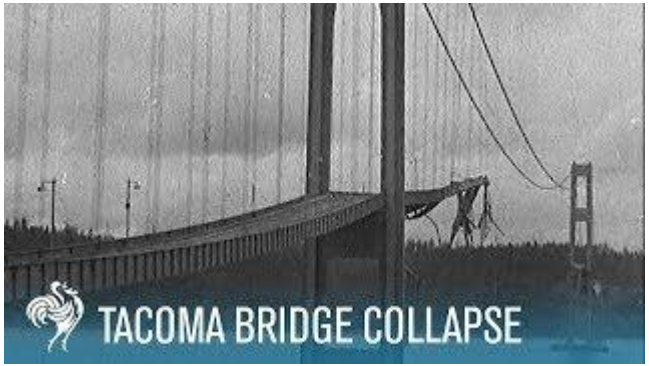
Vibrations



- Natural Frequency and Resonance
- Damping
- Avoiding Resonance & Isolation

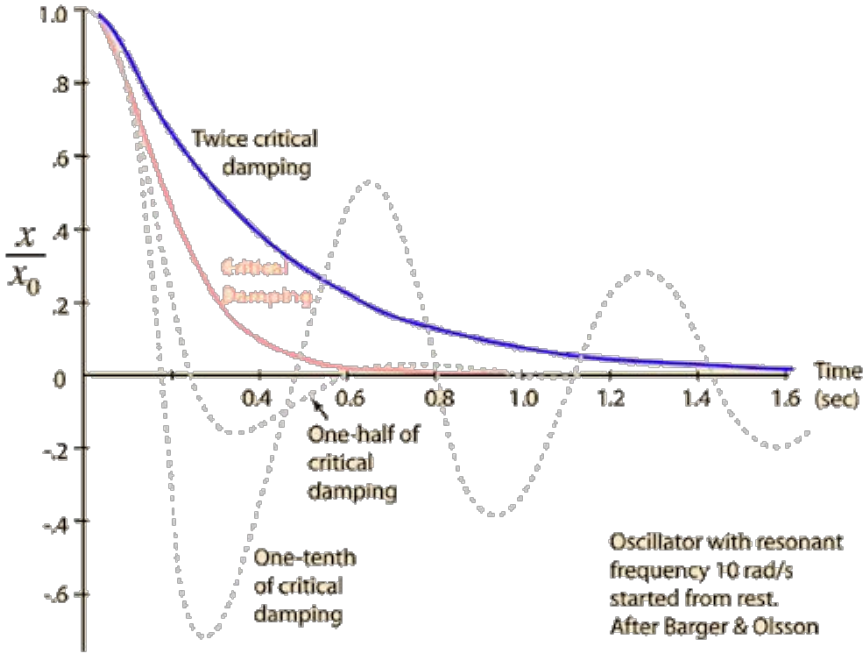
Natural Frequency and Resonance

- Frequency at which a system oscillates when disturbed and left to vibrate freely with no external force
- Resonance occurs when an external forcing frequency matches the natural frequency, causing amplitude to grow large



Damping

- Dissipation of vibrational energy, typically as heat, which reduces oscillation amplitude over time
- Three types
  - Underdamped: oscillates with decreasing amplitude before settling (most common)
  - Critically damped: returns to equilibrium fastest without oscillating (ideal for many control systems)
  - Overdamped: returns to equilibrium slowly, no oscillation



Avoiding Resonance & Isolation

- Best practice: shift the system’s natural frequency away from the expected operating/forcing frequency by changing mass or stiffness
- Add damping to reduce amplitude
- Vibration isolation via special mounts or isolators between vibration source and structure



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Thermodynamics

Thermodynamics



- Four Laws of Thermodynamics
- Thermodynamic Systems
- Thermodynamic Processes
- Heat vs Temperature
- Heat vs Work vs Internal Energy vs Entropy vs Enthalpy

Four Laws of Thermodynamics



- Zeroth Law: if system A is in equilibrium with system B, and system B is in equilibrium with system C, then system A is in equilibrium with system C
- First Law: energy cannot be created or destroyed, only transformed from one form to another. The total energy of an isolated system stays constant
- Second Law: the total entropy of an isolated system always increases over time in any spontaneous process (heat cannot spontaneously flow from a colder body to a warmer body)
- Third Law: as a system's temperature approaches absolute zero, the entropy of a perfect, crystalline substance approaches zero

Thermodynamic Systems



- Open Systems
  - Mass, heat, and work can flow in and out of a system
  - Boiling kettle b/c steam and heat escape the system
- Closed Systems
  - Constant mass, but heat and work can pass through the boundary
  - A sealed metal tin can be heated and cooled, but its contents cannot escape
- Isolated Systems
  - No heat, work, or mass can cross the system boundary
  - A well-insulated thermos holding a hot liquid is a close example (truly perfect isolated systems are exceedingly rare in reality)

Thermodynamic Processes



- Isothermic: constant temperature
- Isochoric: constant volume
- Isobaric: constant pressure
- Adiabatic: no heat transfer between system and environment
- Reversible: direction of process can be reversed by small changes in some properties (e.g. melting ice at 0° C)
- Isentropic: adiabatic and reversible

Heat vs Temperature



Heat:

- The transfer of thermal energy between objects due to a temperature difference
- Represents energy in transit
- An object contains thermal energy, but it does not “possess” heat until that energy moves

Temperature:

- A measure of the average kinetic energy of particles in a substance
- Does not depend on the amount of material present
- Tells you how hot or cold an object is

Heat vs Work vs Internal Energy vs Entropy vs Enthalpy



- Heat (Q)
  - The transfer of thermal energy between objects due to a temperature difference (path)
- Work (W)
  - The transfer of energy through ordered, macroscopic motion that does not rely on a temperature difference (path)
- Internal Energy (U)
  - Total microscopic energy stored inside a system (K+P) (state)
- Entropy (S)
  - Measure of disorder, randomness, or unavailability of a system's energy to do useful work. Dictates the direction of natural processes under the second law of thermodynamics
- Enthalpy (H)
  - Represents the total heat content of a system at constant pressure (U+W) (state)



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Fluids

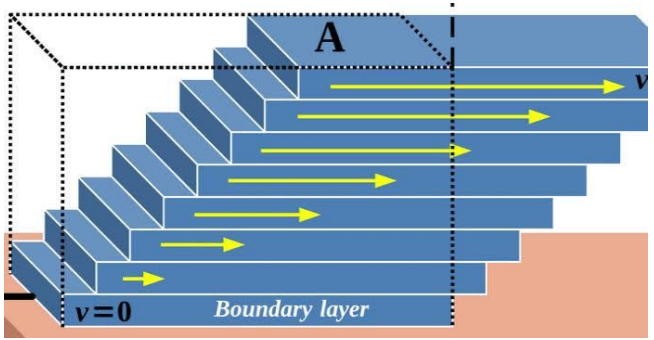
Fluids

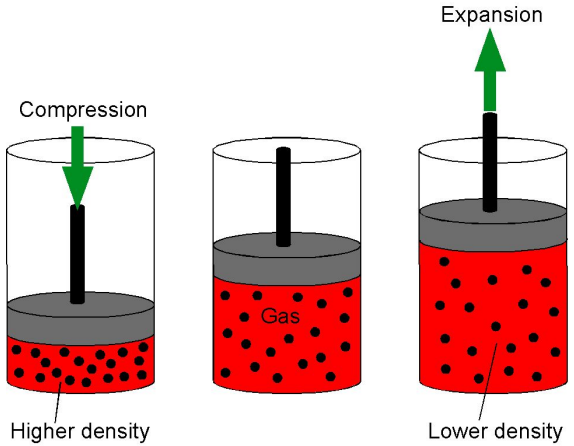


- What is a fluid?
- Viscosity
- Continuity Equation
- Bernoulli’s Principle
- Reynolds Number
- Losses

What is a fluid?

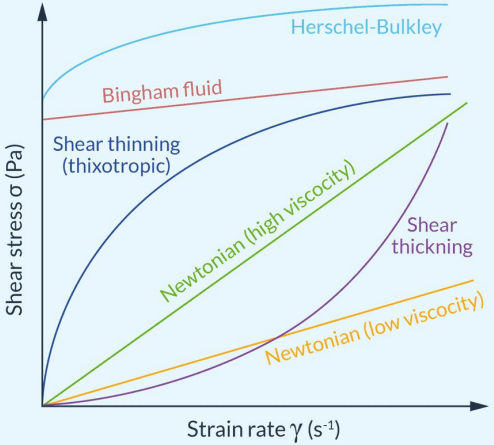
- A fluid is a substance that continuously deforms under shear stress
- Liquids are incompressible and have fixed volume
- Gases are compressible and fill their containers
- No-slip condition





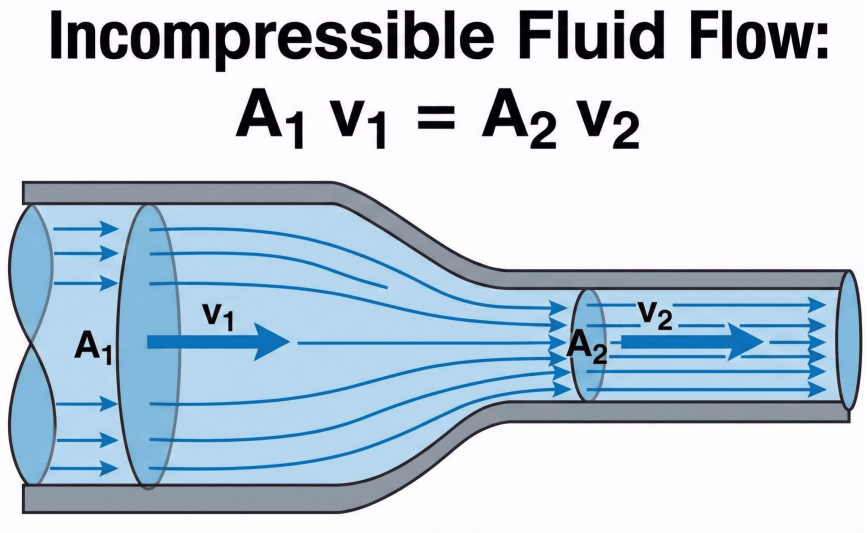
Viscosity

- Viscosity is a fluid's resistance to shear/flow; internal friction
- Dynamic viscosity (v) measures a fluid's internal resistance to flow under an applied force
- Kinematic viscosity (μ) measure resistive flow under the influence of gravity alone
- Newtonian fluids vs non-Newtonian fluids: shear rate vs viscosity
- Define "inviscid"



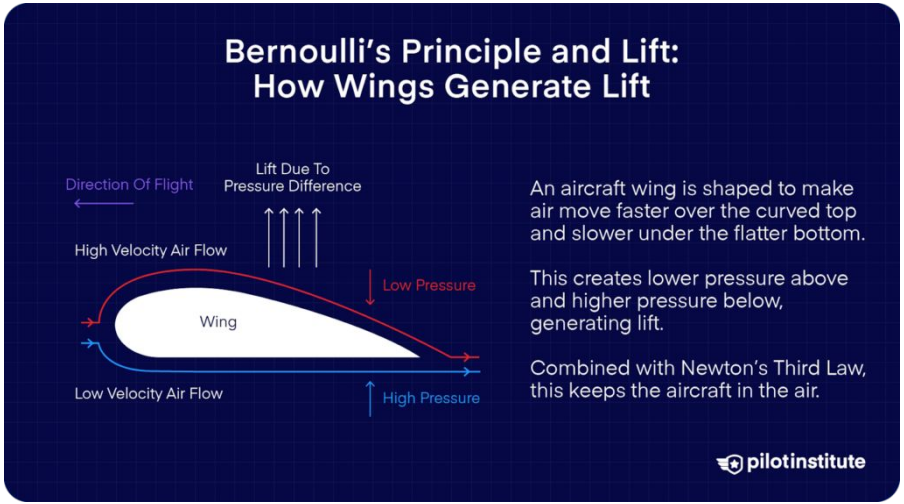
Continuity Equation

- Conservation of mass in a flowing fluid
- $A_1V_1=A_2V_2$ (incompressible flow)
- As pipe area decreases, velocity increases



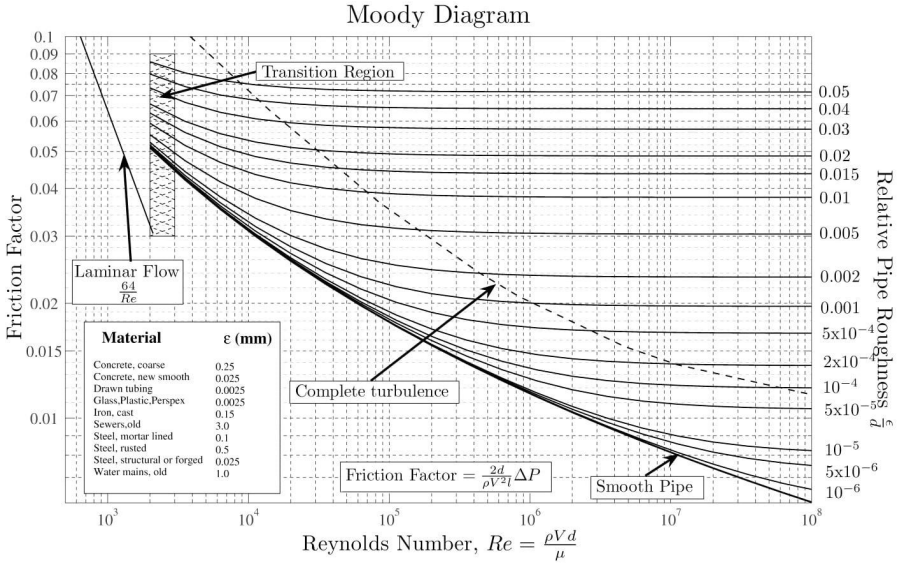
Bernoulli’s Principle

- Conservation of energy along a streamline (pressure + kinetic + potential energy = constant)
- $P + \frac{1}{2}\rho V^2 + \rho g h = \text{constant}$
- Assumes inviscid, incompressible, and steady flow along a streamline



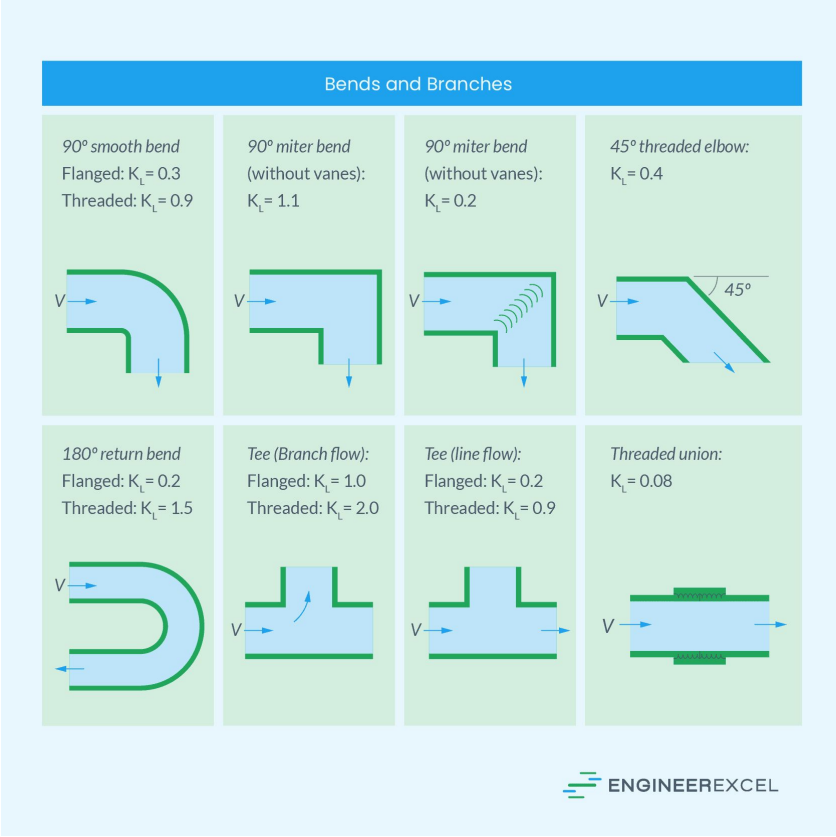
Reynolds Number

- Ratio of internal forces to viscous forces
- Re=ρVD/μ
- Predicts flow regime (estimates, varies from source to source)
  - Re < 2300 Laminar
  - Re > 4000 Turbulent
  - In between = transitional
- Determines friction factor, losses, mixing behavior



Losses

- Major losses: friction along pipe length, depends on Re and pipe roughness
- Minor losses: fittings, valves, bends, sudden expansions/contractions
- Losses irreversibly convert mechanical energy to heat



Wrap-Up



- Breathe
- Take time to think; don’t rush answer
- Verbalize your thought process
- Don’t stress too much :)
- Interview questions



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ANSYS (FEA) I Foundations

Next Week



“Louis, I think this is the beginning of a beautiful friendship”

Rick (Casablanca 1942)

Incubator

Attendance

Replace